$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array}\begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 1, & 1, & 64, & 64, & 64 \\ 1, & 1, & 30, & 64, & 64 \\ 1, & 60, & 32, & 32, & 32 \\ 1, & 120, & 16, & 16, & 16 \\ 1, & 240, & 8, & 8, & 8 \\ 1, & 320, & 4, & 4, & 4 \end{bmatrix} \end{array} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ \\ \end{bmatrix} \quad \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 1, & 320, & 4, & 4, & 4 \\ 1, & 240, & 8, & 8, & 8 \\ 1, & 120, & 16, & 16, & 16 \\ 1, & 60, & 32, & 32, & 32 \\ 1, & 1, & 30, & 64, & 64 \\ 1, & 1, & 64, & 64, & 64 \end{bmatrix} \end{array} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ \\ \end{bmatrix}$$

$$\frac{u_1+v_1}{2}, \frac{u_2+320v_2}{2}, \frac{64u_3+4v_3}{2}, \frac{64u_4+4v_4}{2}, \frac{64u_5+4v_5}{2}$$

$$\frac{u_1+v_1}{2}, \frac{u_2+240v_2}{2}, \frac{32u_3+8v_3}{2}, \frac{64u_4+8v_4}{2}, \frac{64u_5+8v_5}{2}$$

$$\frac{u_1+v_1}{2}, \frac{60u_2+120v_2}{2}, \frac{32u_3+16v_3}{2}, \frac{32u_4+16v_4}{2}, \frac{32u_5+16v_5}{2}$$

$$\frac{u_1+v_1}{2}, \frac{120u_2+60v_2}{2}, \frac{16u_3+32v_3}{2}, \frac{16u_4+32v_4}{2}, \frac{16u_5+32v_5}{2}$$

$$\frac{u_1+v_1}{2}, \frac{240u_2+v_2}{2}, \frac{8u_3+30u_3}{2}, \frac{8u_4+64v_4}{2}, \frac{8u_5+64v_5}{2}$$

$$\frac{u_1+v_1}{2}, \frac{320u_2+v_2}{2}, \frac{4u_3+64v_3}{2}, \frac{4u_4+64v_4}{2}, \frac{4u_5+16v_3}{2}$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 32 & 32 & 32 \\ \frac{1}{2} & \frac{1}{2} & 16 & 32 & 32 \\ \frac{1}{2} & 30 & 16 & 16 & 16 \\ \frac{1}{2} & 60 & 8 & 8 & 8 \\ \frac{1}{2} & 120 & 4 & 4 & 4 \\ \frac{1}{2} & 160 & 2 & 2 & 2 \end{bmatrix}$$

Jacobian_U

D_u

$$\begin{bmatrix} \frac{1}{2} & 160 & 2 & 2 & 2 \\ \frac{1}{2} & 120 & 4 & 4 & 4 \\ \frac{1}{2} & 60 & 8 & 8 & 8 \\ \frac{1}{2} & 30 & 16 & 16 & 16 \\ \frac{1}{2} & \frac{1}{2} & 15 & 32 & 32 \\ \frac{1}{2} & \frac{1}{2} & 32 & 32 & 8 \end{bmatrix}$$

Jacobian_V

Dr

u

$$\begin{bmatrix} 1 & 4 & 256 & 256 \\ 1 & 64 & 256 & 256 \\ 1 & 128 & 128 & 128 \end{bmatrix}$$

v

$$\begin{bmatrix} 128 & 128 & 128 \\ 64 & 256 & 256 \\ 1 & 256 & 256 \end{bmatrix}$$

$$\frac{u_1 + v_1}{2} \quad \frac{4u_2 + 128v_2}{2} \quad \frac{256u_3 + 128v_3}{2} \quad \frac{256u_4 + 128v_4}{2}$$

$$\frac{u_1 + v_1}{2} \quad \frac{64u_2 + 64v_2}{2} \quad \frac{256u_3 + 256v_3}{2} \quad \frac{256u_4 + 256v_4}{2}$$

$$\frac{u_1 + v_1}{2} \quad \frac{128u_2 + v_2}{2} \quad \frac{128u_3 + 256v_3}{2} \quad \frac{128u_4 + 256v_4}{2}$$

D_u

$$\begin{bmatrix} \frac{1}{2} & 2 & 128 & 128 \\ \frac{1}{2} & 32 & 128 & 128 \\ \frac{1}{2} & 64 & 64 & 64 \end{bmatrix}$$

D_v

$$\begin{bmatrix} \frac{1}{2} & 64 & 64 & 64 \\ \frac{1}{2} & 32 & 128 & 128 \\ \frac{1}{2} & \frac{1}{2} & 128 & 128 \end{bmatrix}$$

$$u = \begin{bmatrix} 1 & 4 & 256 & 256 \\ 1 & 128 & 256 & 256 \\ 1 & 256 & 128 & 128 \\ 1 & 512 & 64 & 64 \\ 1 & 1024 & 32 & 32 \end{bmatrix} \qquad v = \begin{bmatrix} 1 & 1024 & 32 & 32 \\ 1 & 512 & 64 & 64 \\ 1 & 256 & 128 & 128 \\ 1 & 128 & 256 & 256 \\ 1 & 4 & 256 & 256 \end{bmatrix}$$

$$+$$

$$\begin{bmatrix} \frac{u_1+v_1}{2}, & \frac{4u_2+1024v_2}{2}, & \frac{256u_3+32v_3}{2}, & \frac{256u_4+32v_4}{2} \\ \frac{u_1+v_1}{2}, & \frac{128u_2+512v_2}{2}, & \frac{256u_3+64v_3}{2}, & \frac{256u_4+64v_4}{2} \\ \frac{u_1+v_1}{2}, & \frac{256u_2+256v_2}{2}, & \frac{128u_3+128v_3}{2}, & \frac{128u_4+128v_4}{2} \\ \frac{u_1+v_1}{2}, & \frac{512u_2+128v_2}{2}, & \frac{64u_3+256v_3}{2}, & \frac{64u_4+256v_4}{2} \\ \frac{u_1+v_1}{2}, & \frac{1024u_2+4v_2}{2}, & \frac{32u_3+256v_3}{2}, & \frac{32u_4+256v_4}{2} \end{bmatrix} \qquad \begin{bmatrix} 0 & 1024 & 0 & 0 \\ 0 & 1024 & 0 & 0 \\ 0 & 512 & 0 & 0 \\ 0 & 256 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$=$$

$$\begin{bmatrix} 1 & 2048 & 32 & 32 \\ 1 & 1536 & 64 & 64 \\ 1 & 768 & 128 & 128 \\ 1 & 384 & 256 & 256 \\ 1 & 4 & 256 & 256 \end{bmatrix}$$

D_4

$\frac{1}{2}$	1	128	128
$\frac{1}{2}$	64	128	128
$\frac{1}{2}$	128	64	64
$\frac{1}{2}$	256	32	32
$\frac{1}{2}$	512	16	16

D_5

$\frac{1}{2}$	1024	16	16
$\frac{1}{2}$	768	32	32
$\frac{1}{2}$	384	64	64
$\frac{1}{2}$	192	128	128
$\frac{1}{2}$	2	128	128

Vertical Stack: $\begin{bmatrix} A \\ B \end{bmatrix}$

Since all matrix sets are $m \times n$ in which m and n are consistent in both sets, $m \times 2n$

$\|p+q+\ldots+k\|$ – total_vessel_length_mm

A_0 – fragmentations

It_1 – false loops

θ – bifurcation count

$\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}$ – Euclidean Distance

$$Re = \frac{uL}{\nu} = \frac{\rho u L}{\mu}$$

$$\rho = \frac{\|p + q + \ldots + k\|}{H_0} \leftarrow \text{fragmentations}$$

$$L = \|p + q + \ldots + k\|$$

$$\mu = \frac{\|p\cos\theta + q\sin\theta\|}{\sqrt{\sum_{i=1}^{h}(p_i - q_i)^2}} \qquad \theta = \text{bifurcation count}$$

$$u = \frac{\|p + q + \ldots + k\|}{H_1} \leftarrow \text{loops}$$

$$Re = \frac{\left(\frac{\|p+q+\ldots+k\|}{H_0}\right)\left(\frac{\|p+q+\ldots+k\|}{H_1}\right)\left(\|p+q+\ldots+k\|\right)}{\frac{\|p\cos\theta + q\sin\theta\|}{\sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}}}$$

$$= \frac{\left(\|p+q+\ldots+k\|\right)^3}{H_0 \cdot H_1} \cdot \frac{\sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}}{\|p\cos\theta + q\sin\theta\|}$$

$$= \boxed{\frac{\left(\|p+q+\ldots k\|\right)^3 \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}}{(H_0)(H_1)(\|p\cos\theta + q\sin\theta\|)}}$$

$Re = f_{(\rho, \mu, L, u)} = \frac{\rho u L}{\mu}$ - Level curve

$$\frac{\partial f}{\partial \rho} = \frac{uL}{\mu} \qquad \frac{\partial f}{\partial L} = \frac{\rho u}{\mu} \qquad \rightarrow \frac{uL}{\mu}\hat{i} + \frac{\rho L}{\mu}\hat{j} + \frac{\rho u}{\mu}\hat{k} - \frac{\rho u L}{u^2}\hat{z}$$

normal vector function

$$\frac{\partial f}{\partial u} = \frac{\rho L}{\mu} \qquad \frac{\partial f}{\partial \mu} = -\frac{\rho u L}{u^2} \qquad \text{point}\left(\frac{\sum_{m=1}^{b}\|p_m + q_m + \ldots + k_m\|}{b}, \frac{\|p+q+\ldots+k\|}{\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}}\right)$$

$$0, \quad 0)$$

$$\frac{uL}{\mu}\left(\frac{\sum_{m=1}^{b}\|p+q+\ldots+k\|}{b}\right)\hat{i}+\frac{\rho L}{\mu}\left(\frac{\|p+q+\ldots+k\|}{\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}\right)\hat{j}$$

$$\frac{\left(\frac{\|p+q+\ldots+k\|}{H_1}\right)\left(\|p+q+\ldots+k\|\right)}{\frac{\|p\cos\theta+q\sin\theta\|}{\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}}\cdot\left(\frac{\sum_{m=1}^{b}\|p+q+\ldots+k\|}{b}\right)\hat{i}+\frac{\left(\frac{\|p+q+\ldots+k\|}{H_0}\right)\left(\|p+q+\ldots+k\|\right)}{\frac{\|p\cos\theta+q\sin\theta\|}{\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}}\cdot$$

$$\left(\frac{\|p+q+\ldots+k\|}{\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}\right)\hat{j}$$

$$\frac{(\|p+q+\ldots+k\|)^2\left(\sum_{m=1}^{b}\|p+q+\ldots+k\|\right)\left(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\hat{i}+\frac{(\|p+q+\ldots+k\|)^3\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}\hat{j}$$

Normal vector

$$\frac{(\|p+q+\ldots+k\|)^2\left(\sum_{m=1}^{b}\|p+q+\ldots+k\|\right)\left(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}\right)(x-x_0)}{b(\|p\cos\theta+q\sin\theta\|)}+\frac{(\|p+q+\ldots+k\|)^3\left(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}\right)(y-y_0)}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}$$

Tangent plane

$$\alpha = L\left(\frac{w\rho}{\mu}\right)^{\frac{1}{2}} = f(L, w, \rho, \mu)$$

$$\underbrace{\frac{(\|p+q+\dots+k\|)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{i=1}^{b}(p_i-q_i)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\hat{i} + \frac{(\|p+q+\dots+k\|)^3\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}\hat{j}}_{\text{Normal vector}}$$

$$\underbrace{\frac{(\|p+q+\dots+k\|)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}\right)(x-x_0)}{b(\|p\cos\theta+q\sin\theta\|)}}_{[c]} + \underbrace{\frac{(\|p+q+\dots+k\|)^3\left(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}\right)(y-y_0)}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}}_{[d]}$$

Tangent plane

$f(L, w, \rho, \mu)$ with a constraint of $c(x-x_0)+d(y-y_0)$

$g((x-x_0),(y-y_0))=$

$$\frac{\partial\alpha}{\partial L} = \left(\frac{w\rho}{\mu}\right)^{\frac{1}{2}}$$

$$\frac{\partial\alpha}{\partial w} = \frac{1}{2}\left(\frac{\rho}{\mu}\right)\left(\frac{w\rho}{\mu}\right)^{-\frac{1}{2}} = \frac{1}{2}\left(\frac{\rho}{\mu}\right)\left(\frac{w^{-\frac{1}{2}}\rho^{-\frac{1}{2}}}{\mu^{-\frac{1}{2}}}\right)$$

$$= \frac{1}{2}\left(\frac{\rho}{\mu}\right)\left(\frac{\sqrt{\mu}}{\sqrt{w\rho}}\right)$$

$$= \frac{\rho\sqrt{\mu}}{2\mu\sqrt{w\rho}}$$

$$\frac{\partial \alpha}{\partial \rho} = \frac{1}{2}\left(\frac{\omega}{\mu}\right)\left(\frac{\omega\rho}{\mu}\right)^{-\frac{1}{2}}$$

$$= \frac{\omega\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}$$

$$\frac{\partial \alpha}{\partial \mu} = -\frac{1}{2}\left(\frac{\omega\rho}{\mu^2}\right)\left(\frac{\omega\rho}{\mu}\right)^{-\frac{1}{2}}$$

$$= -\left(\frac{\omega\rho}{2\mu^2}\right)\left(\frac{\sqrt{\mu}}{\sqrt{\omega\rho}}\right)$$

$$= \frac{-\omega\rho\sqrt{\mu}}{2\mu^2\sqrt{\omega\rho}}$$

$$\left(\frac{\omega_p}{\mu}\right)^{\frac{1}{2}}\hat{i} + \left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega_p}}\right)\hat{j} + \left(\frac{\omega\sqrt{\mu}}{2\mu\sqrt{\omega_p}}\right)\hat{k} - \left(\frac{\omega_p\sqrt{\mu}}{2\mu^2\sqrt{\omega_p}}\right)\hat{i} =$$

$$\lambda \cdot \left[(c + (x - x_0))\hat{i} + (d + (y - y_0))\hat{j}\right]$$

$$\downarrow$$

$$(c\lambda + (x - x_0)\lambda) + (d\lambda + (y - y_0)\lambda)$$

$$\left(\frac{\omega_p}{\mu}\right)^{\frac{1}{2}} = c\lambda + (x - x_0)\lambda \rightarrow \frac{\left(\frac{\omega_p}{\mu}\right)^{\frac{1}{2}} - c\lambda}{\lambda} = (x - x_1)$$

$$\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega_p}}\right) = d\lambda + (y - y_0)\lambda$$

$$\downarrow$$

$$\frac{\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega_p}}\right) - d\lambda}{\lambda} = y - y_0$$

$$d\left[\frac{\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}\right)-d\lambda}{\lambda}\right]+c\left[\frac{\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}}-c\lambda}{\lambda}\right]=L\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}}$$

$$\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}\,\lambda}-d^2+\frac{c\sqrt{\omega\rho}}{\sqrt{\mu}\lambda}-c^2=\frac{L\sqrt{\omega\rho}}{\sqrt{\mu}}$$

$$\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}\lambda}+\frac{c\sqrt{\omega\rho}}{\sqrt{\mu}\lambda}=c^2+d^2+\frac{L\sqrt{\omega\rho}}{\sqrt{\mu}}$$

$$\frac{1}{\lambda}\left(\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}+\frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}\right)=c^2+d^2+\frac{L\sqrt{\omega\rho}}{\sqrt{\mu}}$$

$$\frac{1}{\lambda}=\frac{c^2+d^2+\frac{L\sqrt{\omega\rho}}{\sqrt{\mu}}}{\left(\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}+\frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}\right)}$$

$$\lambda=\frac{\left(\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}+\frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}\right)}{c^2+d^2+\frac{L\sqrt{\omega\rho}}{\sqrt{\mu}}}$$

$$\frac{\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}} - c\lambda = (x - x_1)}{\lambda} \qquad \frac{\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}\right) - d\lambda = y - y_1}{\lambda} \qquad \lambda = \frac{\left(\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}\right)}{c^2 + d^2 + \frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}$$

$$\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}} - c\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right] = y - y_0$$

$$\frac{}{\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]}$$

$$\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}\right) - d\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \frac{\sqrt{\omega\rho}}{\mu}}\right] = x - x_0$$

$$\frac{}{\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{c\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]}$$

$$\frac{\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}} - c\lambda = (x - x_0)}{\lambda} \qquad \frac{\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}\right) - d\lambda = y - y_0}{\lambda}$$

$$\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}} - c\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + c\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right] = q - y_0$$

$$\left[\frac{\frac{d\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + c\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}{c^2 + d^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]$$

$$\underbrace{\frac{(\|p+q+\dots+k\|)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{i=1}^{n}(p_i-p_j)^2}\right)(x-x_0)}{b\,(\|p\cos\theta+q\sin\theta\|)} + \frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)(y-y_0)}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-p_j)^2}}}_{\text{tangent plane}}$$

[c] [d]

$$\left(\frac{\omega\rho}{\mu}\right)^{\frac{1}{2}} - \left[\frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{i=1}^{n}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]\left[\frac{\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)\sqrt{\omega\rho}}{b\,(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu}}}{\left[\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]$$

$$= q - y_0$$

$$\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}}\right) - \left[\frac{\left(\|p+q+\dots+k\|\right)^3\left(\sqrt{\sum_{i=1}^{n}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-p_j)^2}\right)}\right]\left[\frac{\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)\sqrt{\omega\rho}}{b\,(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu}}}{\left[\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]$$

$$= x - x_0$$

$$C.P.\ (x - x_0,\ y - y_0) = f\left(\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} - \left[\frac{\left(\|p+q+\dots+k\|\right)^3\left(\sqrt{\sum_{i=1}^{n}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-p_j)^2}\right)}\right]\left[\frac{\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)\sqrt{\omega\rho}}{b\,(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu}}}{\left[\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right],\ \frac{\left(\frac{\rho}{2\mu}\right)\sqrt{\mu}}{\sqrt{\omega\rho}} - \left[\frac{\left(\|p+q+\dots+k\|\right)^3\left(\sqrt{\sum_{i=1}^{n}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-p_j)^2}\right)}\right]\left[\frac{\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{(\|p\cos\theta+q\sin\theta\|)\left(\sum_{j=1}^{a}\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}\frac{\rho\sqrt{\mu}}{2\mu\sqrt{\omega\rho}} + \frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)\sqrt{\omega\rho}}{b\,(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu}}}{\left[\frac{(\|p+q+\dots+k\|)^2\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{\left(\|p+q+\dots+k\|\right)^2\left(\sum_{m=1}^{b}\|p+q+\dots+k\|\right)\left(\sqrt{\sum_{m=1}^{b}(p_i-p_j)^2}\right)}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \angle\frac{\sqrt{\omega\rho}}{\sqrt{\mu}}}\right]\right)$$

$$f\left(\frac{P\sqrt{\mu}}{2\lambda\sqrt{\varepsilon\rho}} - \frac{\left[\frac{(\rho+q+\ldots+k)^2(\sum_{i=1}^{b}\|p+q+\ldots+k\|)(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-q_i)^2}}\right]\left[\frac{(\rho+q+\ldots+k)^3(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})^2}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}} + \frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}}\right]}{\left[\frac{(\rho+q+\ldots+k)^2\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2} + \angle\frac{\sqrt{\varepsilon\rho}}{\sqrt{\mu}}, \frac{P\sqrt{\mu}}{2\lambda\sqrt{\varepsilon\rho}} - \frac{\left[\frac{(\rho+q+\ldots+k)^2(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-q_i)^2}}\right]\left[\frac{(\rho+q+\ldots+k)^3(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})^2}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}} + \frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}}\right]}{\left[\frac{(\rho+q+\ldots+k)^2\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2} + \angle\frac{\sqrt{\varepsilon\rho}}{\sqrt{\mu}}\right)$$

$$=$$

$$\underbrace{\frac{(\|p+q+\ldots+k\|)^2(\sum_{i=1}^{b}\|p+q+\ldots+k\|)(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})}{b(\|p\cos\theta+q\sin\theta\|)}(x-x_0)}_{} + \underbrace{\frac{(\|p+q+\ldots+k\|)^3(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})^2}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-q_i)^2}}(y-y_0)}$$

where $(x-x_0) =$ $\frac{P\sqrt{\mu}}{2\lambda\sqrt{\varepsilon\rho}} - \frac{\left[\frac{(\rho+q+\ldots+k)^2(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})}{(\|p\cos\theta+q\sin\theta\|)\sum_{j=1}^{a}\sqrt{\sum_{i=1}^{b}(p_i-q_i)^2}}\right]\left[\frac{(\rho+q+\ldots+k)^3(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})^2}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}} + \frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}}\right]}{\left[\frac{(\rho+q+\ldots+k)^2\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2 + \left[\frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)}\right]^2} + \angle\frac{\sqrt{\varepsilon\rho}}{\sqrt{\mu}}$

and $(y-y_0) =$ $\frac{\left(\frac{3\lambda\rho}{\mu}\right)^{\frac{1}{2}}\left[(\|p+q+\ldots+k\|)^2(\sum_{i=1}^{b}\|p+q+\ldots+k\|)(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})\right]}{b(\|p\cos\theta+q\sin\theta\|)}\left[\frac{(\rho+q+\ldots+k)^3(\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2})^2}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}} + \frac{(\rho+q+\ldots+k)^2\sin\theta\sqrt{\sum_{i=1}^{n}(p_i-q_i)^2}}{b(\|p\cos\theta+q\sin\theta\|)\sqrt{\mu\varepsilon}}\right]$